

meesho

SEASON 3

DICE CHALLENGE

REDUCING RTO:
GETTING MORE
ORDERS
DELIVERED

**BUSINESS
TRACK**

The Landscape

India's e-commerce logistics network delivers to some of the most diverse addresses in the world - from dense city lanes to villages with no street names. Cash on Delivery remains the preferred payment mode in smaller towns, and most deliveries are handled by gig workers riding two-wheelers, managed by small local delivery partners rather than large corporate fleets.

Within this setup, one problem quietly costs the industry more than almost any other: orders that are shipped, travel the entire delivery network, but are never delivered - and instead travel all the way back to the seller. This is called RTO (Return to Origin). Every major logistics player in India - whether serving Meesho, Flipkart, Amazon, or Myntra - treats reducing RTO as one of its most important priorities, because it affects cost, customer trust, and how much sellers are willing to sell online.

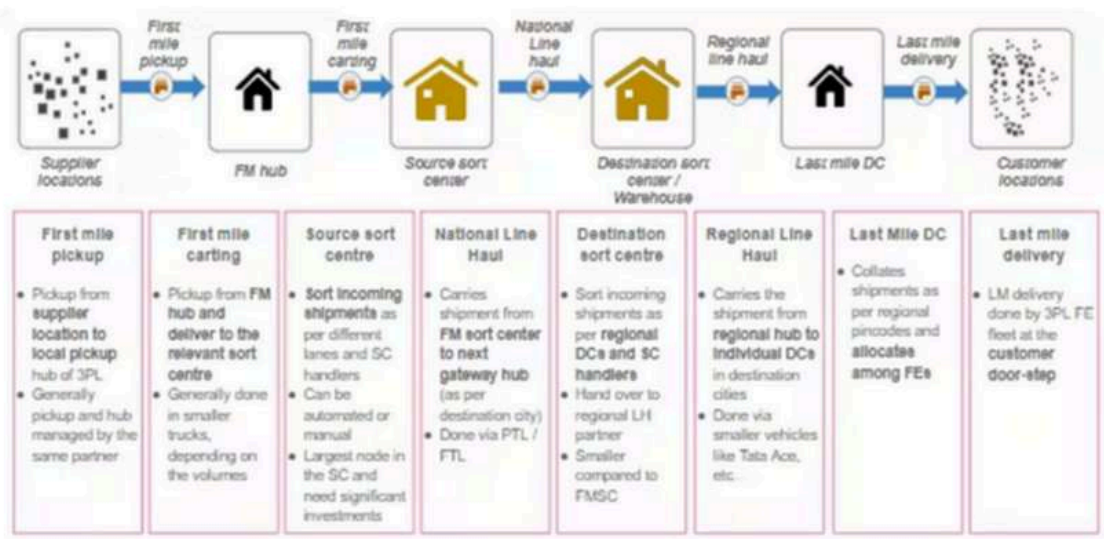
Introduction to Valmo

Valmo is Meesho's in-house logistics platform. The name blends 'Value' and 'Movement,' reflecting its focus on building logistics capabilities in collaboration with regional partners. Valmo works with local delivery entrepreneurs to strengthen connectivity across different parts of the country. The aim is to build a tech-enabled logistics network that brings structure, reliability, and efficiency to the movement of goods. A parcel typically moves through a pickup point, one or two sorting hubs, and finally a local delivery hub close to the customer, from where a delivery rider carries it to the doorstep.

Nodes in Valmo's Supply Chain:

- **First Mile Pickup Hub:** This is the first stage where shipments are collected from sellers or suppliers. Local logistics partners typically manage these hubs, ensuring that goods from different locations are consolidated efficiently for the next step in the network.
- **Source Sort Center:** After pickup, shipments are moved to a source sort center. Here, goods are aggregated and systematically sorted based on their onward destinations. The process helps in streamlining the flow of shipments and assigning them to appropriate long-haul routes.

- **Destination Sort Center:** Situated closer to delivery regions, this facility functions like the source sort center but on the receiving side. Shipments arriving from national line-haul routes are unloaded, organized, and re-sorted according to smaller local delivery zones. This step prepares them for efficient distribution in the final stage.
- **Last Mile Delivery Center:** The last node in the chain, responsible for dispatching shipments to customers. This is the most execution-heavy part of the logistics process, involving navigation of local routes, multiple drop-offs, and timely deliveries. The Local delivery hubs are usually run by small entrepreneurs (who manage riders and hub operations), while the riders themselves are gig workers who are paid per successful delivery. When a delivery doesn't happen and the parcel has to be sent back to the seller, that shipment is marked RTO — and Valmo bears the cost of the failed journey, both ways.



Objective

“Identify the biggest reasons behind failed deliveries (RTO) on Valmo, and design a strategy to meaningfully bring this number down without hurting delivery cost, rider earnings, or the speed and ease of ordering for customers.”

Your solution should look at the problem end-to-end — from the moment an order is placed to the moment it is delivered or returned or marked as an RTO and should pay particular attention to how delivery attempts are made, how riders and customers communicate, how orders sometimes end up at the wrong delivery hub, how address quality affects outcomes, and what could be done with orders that customers refuse to accept.

Expected Benefits

- **Lower delivery costs:** every order saved from returning avoids paying for a failed trip and a reverse trip.
- **Higher customer trust:** a customer who receives their order on time is far more likely to shop online again.
- **Better seller experience:** sellers get paid faster and don't have inventory stuck travelling back and forth.
- **A more efficient network:** hubs and riders spend less time on deliveries that were never going to succeed

Challenges

RTO is not caused by one thing — it is the sum of many small failures across the delivery journey. As you study this problem, you will likely run into challenges such as:

- **Delivery attempts aren't always what they seem.** Not every shipment marked “delivery attempted” by the last mile delivery rider/partner was genuinely attempted. Understanding why, and how to tell a real attempt from one that wasn't, is a real challenge.
- **Riders and customers often can't reach each other.** A rider may be at the right address but unable to reach the customer by phone - and the delivery fails as a result.
- **Customer addresses are often unclear.** Many addresses rely on descriptions and nearby landmarks rather than a fixed structure, which makes it harder for Valmo to route the order to a delivery hub as well as for a new rider to locate.
- **Orders sometimes land at the wrong delivery hub.** An order meant for one part of a city where the customer is located can get routed to a hub that doesn't serve that area, delaying delivery and increasing the chance of failure.
- **Customers refuse orders for many different reasons.** Cash on Delivery makes it easy for a customer to change their mind at the doorstep and refuse to receive the order
- **Refused orders from customers travel all the way back to the sellers** - while incurring double the logistics cost. Is there a way to reduce this reverse logistics cost for orders which are marked RTO?

- **Rider Intent** - Riders are currently paid the same fixed rate for all successful deliveries, for certain specific orders - riders have a lower intent leading to RTO.
- **India is not one market.** A solution that works in a dense city neighbourhood may not work in a small town or village, given differences in address quality, network reach, and rider availability.

Required Output

Prepare a presentation summarising your findings and recommendations. It should be clear, concise, and visually engaging.

Round 1: 3-Slider Submission

1. Map out the biggest reasons behind RTO that you have identified, with rough sizing of each.
2. Share the framework you used to prioritise which problems to solve first.
3. Identify one quick-win idea you would take up in your first 30 days, with a short rationale.

Round 2: Detailed Deck Submission (6–10 slides)

1. Executive summary of your diagnosis and recommendations.
2. A breakdown of where and why RTO happens, backed by research and/or the data pack.
3. Your prioritised solutions, with expected impact and reasoning.
4. A specific proposal for what could be done with refused/undelivered orders instead of sending them all the way back.
5. A 30-60-90 day plan to test your top idea(s), including how you would measure success.
6. Key risks in implementation, and how you would guard against them.

Evaluation Criteria

3-Slider Submission Round

- Extent of Research - Methods, Sources, Coverage
- Quality of Insights
- User-First Approach
- Problem-First Mindset

Detailed Deck Submission

- Quality of Research - Methods, Sources, Coverage
- Depth of Analysis
- Innovativeness of Recommendations
- 10x and Long-term Thinking
- Feasibility of Implementation
- Presentation Skills

Guidelines / How to Score Better on This Case

- Speak to real people wherever possible — delivery riders, hub operators, or COD shoppers (from any platform, not just Meesho) — to ground your insights in reality.
- Play with the Meesho-app to identify flaws, place orders & observe what reasons could lead to an RTO
- Float a short survey to strengthen your primary research on why customers refuse or miss deliveries.
- Look at how other logistics players (Indian or global) have tackled similar problems, and use those as reference points.
- Back your prioritisation with simple, clearly-stated sizing — show your assumptions and your working.
- Think about second-order effects: does your idea hold up once riders, hubs, or customers try to work around it?

Deliverables

Design an RTO reduction strategy for Valmo that answers:

1. What are the biggest reasons orders fail to get delivered, and how big is each?
2. Which of these would you fix first, and why?
3. What would you do differently with orders that customers refuse?
4. How would you test your ideas, and how would you know they worked?
5. What could go wrong, and how would you protect against it?

Assumptions & Data Pack

The figures below are illustrative approximations meant to help you reason about scale — use them as a starting point and build on them with your own research and assumptions.

Snapshot: RTO on Valmo

Metric	Approx. value
Cash on Delivery (COD) order share in Valmo	80%
RTO% of COD orders	20%
RTO% of Prepaid Orders	5%
Forward Logistics Cost for a successful delivery	50 INR / Order
Reverse Logistics Cost for an RTO shipment	120 INR / Order

Logistics Cost Split across Nodes:

Logistics Leg	Approx cost per
FM Hub	4
FM Carting	2
FMSC	5
NLH	8
LMSC	5
RLH	5
LMDC	21

Good luck! We look forward to seeing how you tackle this challenge and come up with innovative ideas for Meesho's growth and success.

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Delivery Success by Distance from Hub

Distance from delivery hub	Nearby (~2 km)	Moderate (~5 km)	Far (~10 km+)
Share of orders returned undelivered	15%	17%	22%

Note: orders placed from a first-time or new address, or with an incomplete/unclear address, tend to see a noticeably higher share of failed deliveries than repeat, well-recognised addresses.

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